Oluwatosin Ayeni

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Doubleword, London, UK

Dear Hiring Team,

I am writing to apply for the Member of Technical Staff: Inference Systems role at Doubleword. Building a distributed batch inference platform that handles high load with low latency is exactly the kind of systems problem I find most interesting — and I have built the closest equivalent I could as an independent engineer.

Three achievements directly relevant to what this role requires:

- **Kubernetes LLM serving platform:** I built *vllm-k8s-stack* — production LLM serving on Kubernetes with a vLLM backend, KV-cache-aware request routing, HPA autoscaling on GPU node pools, and a Prometheus/Grafana observability stack. The system was designed around the constraint that KV cache state cannot be migrated mid-request — so routing decisions must account for prefix reuse. This is the core challenge in a batch inference platform. GitHub
- **Database optimisation under concurrent load:** I run Supabase (PostgreSQL) as the primary data layer for multi-tenant production applications — designed schemas for concurrent session workloads, tuned query plans for hot paths, and instrumented with real-time observability. Sub-500 ms end-to-end latency under bursty concurrent load is a constraint I have engineered around directly.
- **H100 SLO compliance at scale:** Benchmarked vLLM v0.30 vs SGLang v0.5.20 on 2× H100 NVL 94 GB measuring throughput, TTFT, TPOT, and SLO compliance across concurrency 1–128. Peak: 5,965 tok/s; 100% SLO compliance at concurrency 128. Understanding where a serving system breaks its SLO guarantees is as important as peak throughput. Live results.

I write primarily in Python but am actively building Rust tooling for inference systems — I understand why Rust is the right choice for a low-latency inference platform. I am open to relocating to London; Farringdon 2–3 days/week is workable.

Yours sincerely,

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